

CBE 162: Dynamics and Control of Chemical Processes - Fall 2026

Homework 3

Due date: September 22nd at 11:59 pm

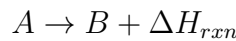
A soft copy must be uploaded on Gradescope. No late homework will be accepted. Please make sure your Name and Student ID Number are in the top-right hand corner of the page. Number all pages in the bottom-right hand corner. Please describe your thought process and state all assumptions made for each question. For numerical answers, place a box around your final answer and remember units.

(1) Object: Predict

Problem 1 (20 points)

A jacketed CSTR is used for an exothermic reaction:

$$r = k \cdot [A] = \frac{d[A]}{dt}$$



The inlet to the CSTR has a flowrate of F_{in} , concentration of $C_{A,in}$, and temperature T_{in} . The fluid volume in this CSTR is constant. We assume F_{in} is constant. The reaction is first order in A, with a rate constant of k . The jacket has a large excess of cooling fluid flowing through it such that the heat transfer can be modeled by the following equation:

$$\dot{Q}_{cool} = U(A) (T_{cool} - T)$$

The cooling temperature T_{cool} can be changed by the operator. We desire to analyze the dynamics of this reactor.

- Identify the MV(s) and the DV(s) in this CSTR.
- Derive the model describing the dynamics of this process. Remember to write out your assumptions.
- Define deviation variables and write the dynamic model in deviation form.
- Consider a step change in T_{cool} with magnitude ΔT_{cool} while all the other DVs and MVs remain constant. Using Laplace transform (that is, without reverting to time domain), determine the new steady-state of the process.
- Determine the time domain response for T for the case given in part (d). Hint: You need to take the inverse Laplace transform.

Problem 2 (15 points)

Find the Laplace transforms of the following functions (inspiration taken from Paul's Math Notes):

(a) $f(t) = 6e^{-2t} + e^{3t} + 4t^4 - 12$

(b) $f(t) = t^{\frac{3}{2}}$

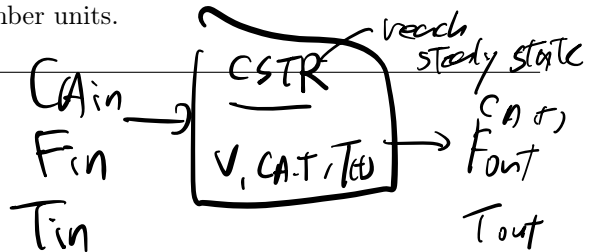
(c) Bonus: $f(t) = t^2 \sin(4t)$

Find the inverse Laplace transforms of the following functions:

(a) $F(s) = \frac{1}{s} - \frac{1}{s-8}$

(b) $F(s) = \frac{e^{-5s}}{s}$

(c) $F(s) = \frac{1}{s^2 + 8s + 7}$



① Identify variable.

② Overall mass balance

Total mass

③ Species mass balance

- Concentration
- mass fraction
- mole fraction

④ if necessary, energy balance

(1) Manipulate variable

Cooling Temperature

T_{cool}

(can be changed by operator)

Perturbance variable

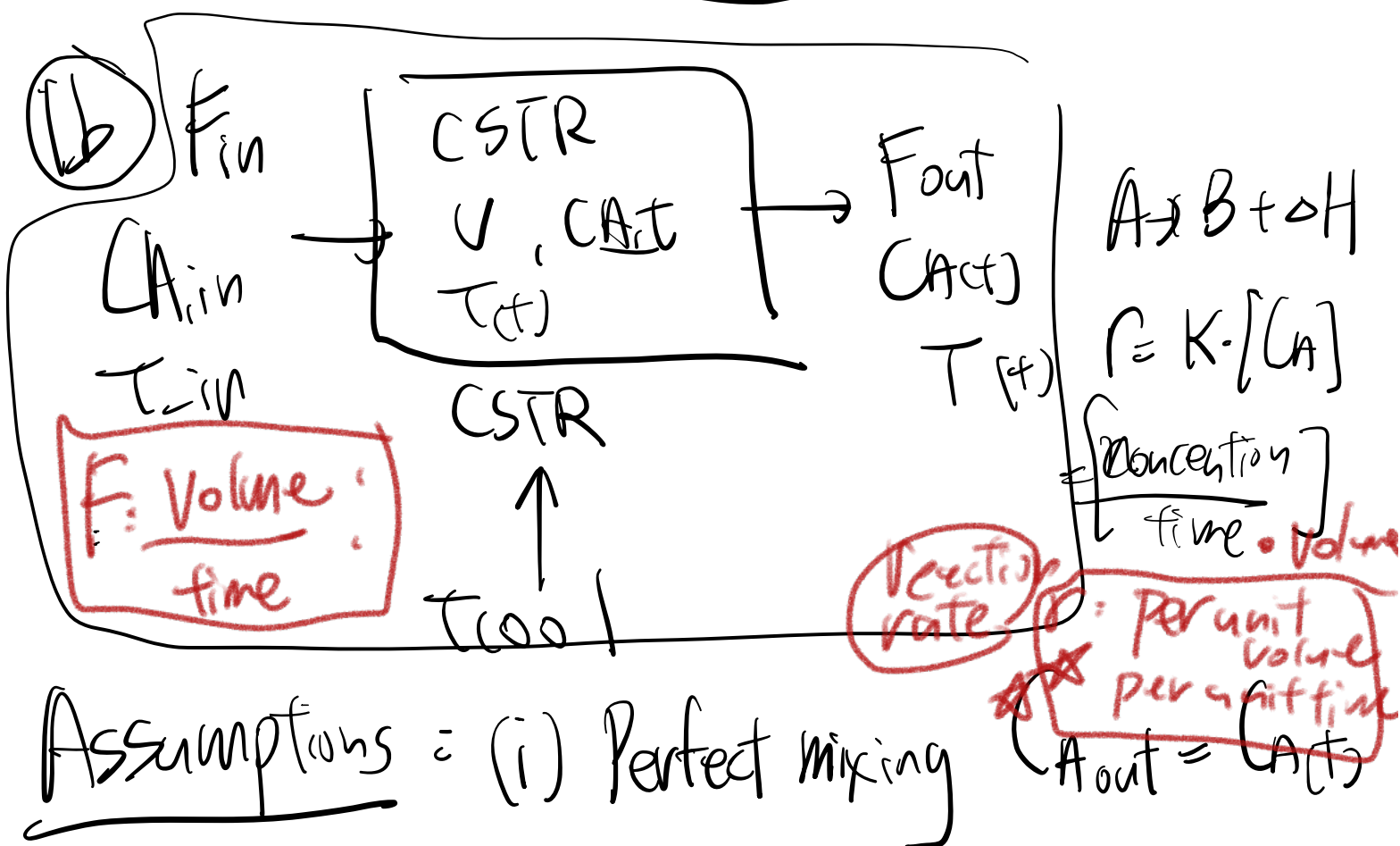
C_{Ain}

A_{in}
(constant)

T_{in}

Something not T_{cool}

$(V = T_{out})_{on}(T)$



(ii) Constant Volume, $V = \text{constant} \rightarrow$ Volumetric Flow rate F

$\Rightarrow$ Volume occupied by liquid

$F_{in} = F = F_{out}$

(iii) constant density (iv) ~~isothermal~~ constant heat capacity (v) K is constant (vi) Adiabatic not heat loss

①

Overall mass balance

$$\frac{dm}{dt} = (F_{in} - F_{out}) \cdot \rho \rightarrow F_{in} = F_{out} = F$$

$$\text{mass} = \rho \cdot V$$

$$\frac{dV}{dt} = F$$

Exothermic

$$\dot{Q} = \cancel{\rho} \cdot H_{rxn} \cdot (K \cdot C_A) V + U \cdot A (T_{cool} - T)$$

heat rate mole enthalpy

②

Species mass balance

$$\frac{d(C_A \cdot V)}{dt} = \cancel{\rho} \cdot V + F \cdot C_A - F_{out} \cdot C_{Aout}$$

also ③

Internal Energy

Energy balance

$$\frac{dU_{sys}}{dt} = F_{in} \cdot \rho \cdot h_{in} - F_{out} \cdot \rho \cdot h_{out} + \dot{Q} \text{ heat rate}$$

Specific enthalpy

$$E = C_p \cdot dT \cdot (V \cdot \rho)$$

mass

$$H = U + PV$$

C_p : specific heat capacity per mass per unit temp / H : Energy

for ②, ③ : assume $\frac{\text{Volume}}{\text{constant}}$, $\frac{\text{pressure}}{\text{fluid}}$

$$dU_{\text{sys}} = dH_{\text{sys}} = (m) C_p \cdot dT = (P \cdot V) C_p \cdot dT$$

$$= d(P \cdot V) \cdot C_p \cdot (T - T_{\text{ref}})$$

$\downarrow$ $\frac{dh}{dt}$

Enthalpy = $h = C_p \cdot dT$ ⑤

$$h_{\text{in}} - h_{\text{out}} = C_p \cdot (T_{\text{in}} - T)$$

How controlled variable change over time: $\frac{dT}{dt} = ?$

$$\frac{dU_{\text{sys}}}{dt} = \frac{d(C_p \cdot dT \cdot (P \cdot V))}{dt} = F \cdot \rho (h_{\text{in}} - h_{\text{out}})$$

$$+ \dot{Q} \Rightarrow F \cdot \rho \cdot [C_p (T_{\text{in}} - T)] - [UA (T_{\text{cool}} - T)]$$

$$+ UA (T_{\text{cool}} - T) \downarrow$$

(b)

$$\frac{dT}{dt} = \frac{F \cdot (T_{in} - T)}{V} - \frac{(\Delta H \cdot k \cdot C_A)}{C_p \cdot P} + \frac{U \cdot A \cdot (T_{cool} - T)}{C_p \cdot P \cdot V}$$

#

2 equation, 2 unknown $C_A(t)$, $T(t)$
 $\rightarrow DOF = 0$

(c) Deviation-based model

At steady state
 Concentration

$$\frac{dC_A}{dt} = 0, \frac{dT}{dt} = 0,$$

$C_{Ain}, \bar{C}_A, \dots$

$$\bar{C}_A = 0, \bar{T} = 0$$

$$0 = F(C_{Ain} - \bar{C}_A) - k V C_A$$

$$\bar{C}_A = \frac{F}{F + kV} C_{Ain}$$

(1)

$$\begin{aligned} C_A' &= C_A - \bar{C}_A, & C_{Ain}' &= C_{Ain} - C_{Ain} \\ T' &= T - \bar{T}, & T_{cool}' &= T_{cool} - T_{cool} \end{aligned}$$

subtract
steady
state
as
deviator

Dynamic
Model

balance
original

$$\frac{d\bar{C}_A}{dt} = 0$$

$$\frac{dC_A'}{dt} = \frac{dC_A}{dt} \cdot V - \frac{d\bar{C}_A}{dt} \cdot V = F \cdot (C_{Ain} - C_A) - k \cdot V \cdot C_A$$

now substitute:

$$C_{Ain}' = C_{Ain}' + \bar{C}_{Ain}, \quad C_A' = C_A' + \bar{C}_A$$

$$\left(\frac{dC_A'}{dt} \cdot V \right) = F \cdot [C_{Ain}' + \bar{C}_{Ain} - C_A' - \bar{C}_A] - k \cdot V \cdot (C_A' + \bar{C}_A)$$

from steady state equation
 plug into (7) (8)

$$F(C_{Ain} - \bar{C}_A) - kV\bar{C}_A = 0$$

$$\frac{dC_A}{dt} \cdot V = F(C_{Ain} - C_A) - (k \cdot V - C_A)$$

from (6)

$$\frac{dT}{dt} = \frac{T' + \bar{T}_{ss}}{dt} = \frac{F \cdot (T_{in}' + \bar{T}_{in,ss}) - (T' + T_{ss})}{\rho V} + \left[\frac{1}{C_p \cdot V} \right] \cdot$$

$$F \cdot k(C_{Ain} - \bar{C}_A) \Delta T \cdot V + V \cdot A \cdot [T_{cool}' + \bar{T}_{cool,ss} - (T' + T_{ss})]$$

$\frac{dT_{ss}}{dt} = 0 \rightarrow$ steady state term = 0

$$\left\{ \begin{aligned} &F(T_{in,ss} - T_{ss}) \\ &+ \frac{1}{C_p \cdot V} [k \cdot C_{ss} \cdot V + \\ &\quad V A (T_{cool,ss} - T_{ss})] \end{aligned} \right\} = 0$$

$$\frac{dT}{dt} = \frac{dT}{dt} \text{ (How temp change over time)} = \frac{F - T_{in} - T'}{V} \quad (1)$$

$$t = \frac{1}{C \cdot \rho \cdot V} \left[(K \cdot C_A' \cdot \Delta H \cdot V) + (V \cdot A \cdot (T_{cool}' - T')) \right]$$



1-d) Unit step change of T_{cool}

$$u(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases}$$

$$T_{cool}(t) = T_{cool,ss} + \left[\Delta T_{cool} \cdot u(t) \right]$$

$$T'_{cool} = \Delta T_{cool} \cdot u(t)$$

$$\mathcal{L}(u(t)) = \frac{1}{s}$$

$$L(T'_{cool}) = \frac{\Delta T_{cool}}{s} = T_{cool}(s)$$

for deviation of temperature (T') changing over time but at steady state

$$\frac{dT'}{dt} = \underbrace{F(T'_{in} - T')}_{(T'_{in} \text{ at } ss = 0)} + \frac{1}{C_p V} \left[\underbrace{-k A (T'_{cool} - T)}_{(C' \text{ at } ss = 0)} \right]$$

$$\frac{dT'}{dt} = \frac{F(T')}{V} + \frac{U \cdot A}{C \cdot p \cdot V} \left(\Delta T_{cool} \cdot u(t) - T' \right)$$

$$\text{Laplace} \left(\frac{dT'}{dt} \right) = s \cdot T'(s) \quad (1)$$

$$\text{Laplace} \left(\frac{f(t)}{dt} \right) = s \cdot F(s) - f(0)$$

$$s \cdot T'(s) = \frac{F(-)T'(s)}{V} + \frac{U \cdot A}{C \cdot p \cdot V} \left[\frac{\Delta T_{cool}}{s} - T'(s) \right]$$

→

$$sT'(s) + \frac{U \cdot A}{C \cdot p \cdot V} \cdot T'(s) + \frac{F \cdot T'(s)}{V}$$

$$= \frac{U \cdot A}{C \cdot p \cdot V} \cdot \frac{\Delta T_{cool}}{s}$$

(b)

$$T'(s) = -\frac{U \cdot A}{C \cdot p \cdot V} \cdot \Delta T_{cool}$$

(14)

$$s \cdot \left[s + \frac{UA}{C \cdot p \cdot V} + \frac{F}{V} \right]$$

New steady state
after change of
 T_{cool}

$$\Rightarrow \lim_{t \rightarrow \infty} T'(t)$$

final value theorem
if and only

$$\lim_{t \rightarrow \infty} T'(t) \text{ exist}$$

$$\lim_{t \rightarrow \infty} T'(t) = \lim_{s \rightarrow 0} s \cdot T'(s)$$

for (14) $s \rightarrow 0$

$$= \frac{U \cdot A \cdot \Delta T_{cool}}{C \cdot p \cdot V}$$

$$\frac{UA}{C \cdot p \cdot V} + \frac{F}{V}$$

by defi:

$$\lim_{t \rightarrow \infty} T'(t) = \lim_{t \rightarrow \infty} (T - T_{old,ss}) = \lim_{t \rightarrow \infty} T - T_{old,ss}$$

$$T'(t) = T - T_{ss,old}$$

$$= T_{new,ss} - T_{old,ss}$$

#

1-e

Time domain response of $(T's)$

$$\text{Laplace}^{-1} (T(s)) = ?$$

for

$$T(s), a = \frac{F}{V} + \frac{V_A}{C \cdot P_V}$$

$$\begin{aligned} & L^{-1} \left(\frac{1}{s(s+a)} \right) \\ &= \frac{1}{a} (1 - e^{-at}) \end{aligned}$$

$$L^{-1} (T(s)) = \frac{\frac{-V_A}{C \cdot P_V} \cdot \Delta T(00)}{\frac{F}{V} + \frac{V_A}{C \cdot P_V}} \left(1 - e^{-t \left(\frac{F}{V} + \frac{V_A}{C \cdot P_V} \right)} \right)$$

~~1~~

2 $L(e^{-at}) = \frac{1}{s+a}$ $L(t^n) = \frac{n!}{s^{n+1}}$ 24

(a) $L[f(t)] = 6 \cdot \left(\frac{1}{s+2}\right) + \left(\frac{1}{s-3}\right) + 4\left(\frac{4!}{s^5}\right)$

$$L(1) = \int_0^{\infty} (1) \cdot e^{-st} \cdot dt$$

$$\left[-\frac{1}{s} e^{-st} \right]_0^{\infty}, e^{-s\infty} \rightarrow 0$$

$$= 0 - \left(-\frac{1}{s}\right) = \frac{1}{s}$$

$$-\frac{12}{s}$$

(24x4)

(a): $F(s) = \frac{6}{s+2}$

$$+ \frac{1}{s-3} + \frac{96}{s^5} \sim \frac{12}{s}$$

(b)

$$L[f(t)] = \int_0^{\infty} t^{\frac{3}{2}} \cdot e^{-st} \cdot dt$$

integration by parts

$$\int u \cdot dv = uv - \int v \cdot du$$

$$u = t^{\frac{3}{2}} \quad v = -\frac{1}{s} \cdot e^{-st}$$

$$du = \frac{3}{2} \cdot t^{\frac{1}{2}} \cdot dt \quad dv = e^{-st} \cdot dt$$

$$= \left[t^{\frac{3}{2}} \cdot \left(-\frac{1}{s}\right) \cdot e^{-st} \right]_0^{\infty} - \int_0^{\infty} t^{\frac{1}{2}} \cdot e^{-st} \cdot dt \cdot \left[\frac{-3}{s-2} \right]$$

as $t \rightarrow \infty$?

$$\frac{t^{\frac{3}{2}}}{e^{-st}} = 0$$

grows faster than polynomial
exponential

$$v = \int e^{-st} \cdot dt = \int e^{-st} \cdot \frac{dt}{d^{-st}}$$

$$= \left(-\frac{1}{s}\right) \cdot e^{-st}$$

as $t \rightarrow 0$ $t^{\frac{3}{2}} \cdot e^{-st} = 0$, so

$$\begin{aligned} u &= t^{\frac{3}{2}} & v &= \left(-\frac{1}{s}\right) \cdot e^{-st} \\ du &= \frac{3}{2} \cdot t^{\frac{1}{2}} dt & dv &= e^{-st} \cdot dt \end{aligned}$$

$$L[f(t)] = \frac{3}{2s} \int_0^{\infty} t^{\frac{1}{2}} \cdot e^{-st} dt$$

Integration by parts.

$$L[f(t)] = \left(\frac{3}{2s}\right) \left[-\frac{t^{\frac{1}{2}}}{s} \cdot e^{-st} \right]_0^{\infty} - \left(-\frac{1}{2s}\right) \int_0^{\infty} e^{-st} \cdot t^{-\frac{1}{2}} dt$$

too much..

(b) for non-integer power, do
Gamma-function formula γ

$$\gamma(z) = \int_0^{\infty} x^{z-1} \cdot e^{-x} dx$$

$$\gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

$$\gamma(z+1) = z \cdot \gamma(z)$$

$$f(t) = t^{3/2}$$

$$\boxed{L(t^\alpha) = \frac{\Gamma(\alpha+1)}{s^{\alpha+1}}}$$

$0 < \alpha < 1$

$$L[f(t)] = \frac{\Gamma(\frac{5}{2})}{s^{\frac{5}{2}}} = \frac{\frac{3}{2} \Gamma(\frac{3}{2})}{s^{\frac{5}{2}}}$$

$$= \frac{\frac{3}{2} \cdot \frac{1}{2} \Gamma(\frac{1}{2})}{s^{\frac{5}{2}}} = \frac{\frac{3}{2} \cdot \frac{1}{2} \cdot \sqrt{\pi}}{s^{\frac{5}{2}}} = \boxed{\frac{3\sqrt{\pi}}{4 \cdot s^{\frac{5}{2}}}}$$

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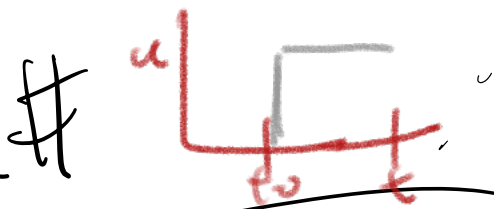
Inverse Laplace ↓

$$L(1) = \frac{1}{s}, \quad L(e^{-at}) = \frac{1}{s+a}$$

2-2a) $f(t) = \text{Laplace}^{-1}(F(s))$

$$= 1 - e^{-st}$$

$u(t-t_0)$: unit step function at t_0



2-2b)

Notice that

$$L(\gamma(t)) = \frac{1}{s}$$

$\gamma(t)$ is unit step function $\begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases}$

Also notice time delay function!

$$L(\gamma(t-a) \cdot f(t-a))$$

$$= e^{-as} \cdot F(s)$$

(where $F(s) = L(f(t))$)

So shift time domain by a ($t-a$)
can be created by $L(f(t)) \cdot e^{-as}$

$$\boxed{e^{-5s} \cdot \frac{1}{s} = \mathcal{L}^{-1} [r(t-5) \cdot f(t-5)]} = \frac{1}{s}$$

also $f(t-5) \cdot f(t) = 1$ because $2(r(t) f(t))$

$\downarrow$

$$\overset{f(s)}{e^{-5s}} \cdot \frac{1}{s} = \mathcal{L}^{-1} [\overset{f(t)}{r(t-5)}]$$

$$f(t) = \begin{cases} 0 & t < 5 \\ 1 & t \geq 5 \end{cases}$$

#

2-2-C

$$F(s) = \frac{1}{s^2 + 8s + 9}$$

$$s^2 + 8s + 9 = (s+1) \cdot (s+9)$$

real,
distinct
roots

$$\boxed{F(s) = \frac{A_1}{s+1} + \frac{A_2}{s+9}}$$

multiply
by
(s+1)(s+9)

$$1 = (s+1) \cdot \alpha_1 + (s+1) \cdot \alpha_2$$

$$\alpha_1 + \alpha_2 = 0, \quad 1 \cdot \alpha_1 + \alpha_2 = 1 \rightarrow \alpha_1 = \frac{1}{6}, \quad \alpha_2 = -\frac{1}{6}$$

$$F(s) = \frac{\frac{1}{6}}{s+1} + \frac{-\frac{1}{6}}{s+1}$$